

OGECHI OKAFOR

AI Engineer

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I design and ship agentic AI systems that go beyond experimentation - multi-agent orchestration pipelines, MCP-integrated tool architectures, and autonomous agents running in production. My background includes four years as a domain analyst across e-commerce, fintech-adjacent, and live events, which means I understand how enterprise data actually behaves and where AI decisions break down at the business level. I translate ambiguous problems into agent-based architectures with clear execution paths, and I instrument everything so I know when something stops working before the business does.

TECHNICAL SKILLS

Agentic Systems	Multi-agent orchestration, single-agent and multi-agent architectures, tool usage, inter-agent communication, autonomous scheduling, agent-to-tool connectivity
MCP & Integration	Model Context Protocol (MCP), custom MCP server design, API integrations, authentication flows, SaaS platform connectors, distributed systems
LLM & Frameworks	OpenAI, Anthropic, LangChain, LangGraph, prompt engineering, structured outputs, function/tool calling, transformer architecture
RAG & Retrieval	Pinecone, ChromaDB, embedding strategies, semantic search, agentic RAG, Ragas evaluation, retrieval quality gates
ML Engineering	XGBoost, Prophet, Sentence Transformer fine-tuning, hard negative mining, feature engineering, MLOps, model evaluation
Backend	Python, FastAPI, PostgreSQL, Redis, RESTful API design, Alembic, SQLAlchemy
Cloud & DevOps	AWS S3, Docker, Railway, Render, GitHub Actions CI/CD
Observability	LangSmith tracing, Ragas quality gates, Mem0 cross-session memory, custom agent instrumentation
Data & BI	SQL, dbt, Airflow, AWS Glue, Tableau, Power BI, Looker

AI ENGINEERING PROJECTS

KIRA — Knowledge Intelligence & Reporting Agent [*Live in Production*]

Multi-agent RAG system with full MCP integration and live evaluation gates.

- Architected a four-agent orchestration pipeline — Research → Analysis → Report → Compliance — with explicit inter-agent handoffs validated via Pydantic. Each agent is independently scoped, testable, and replaceable without breaking the pipeline.
- Implemented the MCP tool protocol for agent-to-tool connectivity, enabling agents to discover and call external tools dynamically without hardcoded integrations. Mem0 provides persistent cross-session memory across the pipeline.
- Dual vector store strategy: ChromaDB in development, Pinecone in production. Benchmarked chunking strategies with Ragas before committing to sentence_window (faithfulness 0.917). Live Ragas quality gates run on every retrieval and generation cycle.

- LangSmith observability across the full pipeline for tracing, debugging, and performance monitoring. FastAPI backend on Render, Streamlit frontend on Streamlit Cloud, AWS S3 for report persistence.

ARIA — Autonomous Repricing & Inventory Agent [Live in Production]

A three-layer decision agent automating ecommerce pricing without human intervention.

- Designed a rules → XGBoost → LLM escalation architecture — each layer handles what it handles best, and escalates only when it lacks confidence. LLM invocation held to 11.1%, reducing API costs by ~90% compared to a naive LLM-first approach.
- XGBoost pricing model trained on 43 real market features from SerpAPI competitor data across 18 products — $R^2 = 0.964$, MAE = \$3.98. Prophet demand forecasting on 157 weeks of Google Trends data with per-category seasonal models.
- FastAPI backend (16 endpoints) deployed on Railway; agent runs on an autonomous hourly schedule. AWS S3 model store with cold-start download and automatic retrain fallback. Full MLOps stack: Docker, GitHub Actions CI/CD, Alembic, PostgreSQL + Redis.

Sentence Transformer Fine-Tune — Stocks & Trading Domain

Domain-adapted embedding model for financial document retrieval.

- Fine-tuned a Sentence Transformer on stocks and trading documents to close the gap between general-purpose embeddings and financial language — where semantically similar terms carry fundamentally different meanings in context.
- Two-round training: round one used anchor/positive pairs with in-batch easy negatives. Round two mined hard negatives — documents close in surface meaning but different in financial context - training the model to push them apart in the boundary regions of vector space where retrieval errors occur.

PROFESSIONAL EXPERIENCE

Data Analyst · Reps & Co

California, USA (Remote) | 2024 – 2025

- Built SQL-backed dashboards tracking ticket demand, pricing, and inventory risk — cut decision time by 35% across purchasing and revenue teams.
- Identified high-value customer segments through behavioral analysis; conversion up 15%, inventory risk down 20%. Findings directly shaped pricing strategy.
- Automated recurring reporting in Python and SQL, reducing manual effort by 30% and opening real-time analytics to cross-functional teams.
- Collaborated with engineering teams to centralize event data from multiple sources, creating a unified analytics framework that improved data quality and accelerated product decision-making.

Lead Data Analyst · Tech4Dev

Lagos, Nigeria (Remote) | 2022 – 2024

- Led full analytics function across data, program, and product teams — owned the pipeline from data validation to stakeholder insight delivery.
- Built predictive models and scenario analysis in Python and SQL; 30% improvement in resource allocation and program optimisation.
- Automated leadership dashboards in Tableau and SQL, cutting manual reporting effort by 40%.

Sales Analyst · HPZ

Lagos, Nigeria | 2019 – 2021

- Built customer behaviour targeting algorithms that converted 45% of seasonal prospects into buyers.
- Improved forecasting accuracy through historical sales analysis, contributing to a 30% lift in B2C sales.

EDUCATION

Computer Science · Federal Polytechnic, Oko